



Review

Tilapia sex determination: Where temperature and genetics meet[☆]J.F. Baroiller^{a,*}, H. D'Cotta^a, E. Bezault^{a,b}, S. Wessels^c, G. Hoerstgen-Schwark^c^a CIRAD, Upr20, Dept. Persyst, Campus International de Baillarguet, F-34398 Montpellier, France^b Institute of Ecology and Evolution, University of Bern and Centre of Ecology, Evolution and Biogeochemistry, EAWAG, CH-6047 Kastanienbaum, Switzerland^c University of Göttingen, Institute of Animal Husbandry and Genetics, Albrecht Thaer-Weg 3, G-37075 Göttingen, Germany

ARTICLE INFO

Article history:

Received 25 August 2008

Received in revised form 20 November 2008

Accepted 20 November 2008

Available online 6 December 2008

Keywords:

TSD

GSD

Selection response

Genes

Wild populations

ABSTRACT

This review deals with the complex sex determining system of Nile tilapia, *Oreochromis niloticus*, governed by the interactions between a genetic determination and the influence of temperature, shown in both domestic and wild populations. Naturally sex reversed individuals are strongly suggested in two wild populations. This can be due to the masculinising temperatures which some fry encounter during their sex differentiation period when they colonise shallow waters, and/or to the influence of minor genetic factors. Differences regarding a) thermal responsiveness of sex ratios between and within Nile tilapia populations, b) maternal and paternal effects on temperature dependent sex ratios and c) nearly identical results in offspring of repeated matings, demonstrate that thermosensitivity is under genetic control. Selection experiments to increase the thermosensitivity revealed high responses in the high and low sensitive lines. The high-line showed ~90% males after 2 generations of selection whereas the weakly sensitive line had 54% males. This is the first evidence that a surplus of males in temperature treated groups can be selected as a quantitative trait. Expression profiles of several genes (*Cyp19a*, *Foxl2*, *Amh*, *Sox9a,b*) from the gonad and brain were analysed to define temperature action on the sex determining/differentiating cascade in tilapia. The coexistence of GSD and TSD is discussed.

© 2008 Elsevier Inc. All rights reserved.

Contents

1. Introduction	30
2. Genetic sex determination of tilapia	31
3. Temperature influences on sex ratios in domestic strains: windows of sexual lability and heritability	31
4. Evidence of temperature-sensitivity in wild populations.	32
5. Temperature action on the sex determining/differentiating cascade in tilapia.	34
6. Coexistence of GSD and TSD in tilapia?	35
Acknowledgements	36
References	36

1. Introduction

Tilapias are currently the second most farmed group of fish (behind carps) with an annual world production of 2.5 million tons (FAO, 2008). This ranking is due to the advantageous aquaculture traits of two tilapia species, *Oreochromis niloticus* and *O. aureus*, and of some hybrids (Baroiller and Toguyeni, 2004). This production relies on all-male (or monosex) populations in order to 1) avoid pond over-

crowding due to their precocious sexual maturity and continuous reproduction, associated to an elaborated parental care and 2) to benefit from male's higher growth-rate (Baroiller and Jalabert, 1989; Beardmore et al., 2001). Currently male monosex populations are produced mainly by androgen treatments. Due to various environmental issues related to hormone use i.e. possible effects of treatment residues on water quality and biodiversity with the growing concerns for food security, finding a sex control alternative based on non-hazardous, consumer and environment-friendly methods represents a major challenge for aquaculture. The best way to obtain all-male populations is through genetic control (Baroiller and Jalabert, 1989; Beardmore et al., 2001). Based on the first data on tilapia sex determination and differentiation, it has been possible to produce

[☆] Contribution associated with the 6th International Symposium on Fish Endocrinology held in June 2008 in Calgary, Canada.

* Corresponding author. Tel.: +33 467593951; fax: +33 467593825.

E-mail address: baroiller@cirad.fr (J.F. Baroiller).

genetically “all-male populations” through the development of YY “supermales” (Baroiller and Jalabert, 1989; Scott et al., 1989). Nevertheless, this approach is unreliable and hampered by the very long procedure of producing and identifying putative YY male individuals (Tessema et al., 2006). Moreover, sex determination has been shown to be more complex than a simple XX/XY monofactorial system.

2. Genetic sex determination of tilapia

Similar to what is seen in most studied teleosts, classical karyotype analysis did not display any dimorphic differences between males and females in Nile tilapia (Majumdar and McAndrew, 1986; Bezault et al., 2001). The sex chromosomes in this species appear to be at an early stage of differentiation (Cnaani et al., 2008). Interspecies hybridization, gynogenesis and progeny testing following sex inversions by hormone treatments, demonstrated that *O. niloticus* has an XX/XY chromosome sex determination similar to mammals (Jalabert et al., 1974; Müller-Belecke and Hörstgen-Schwark, 1995). Similar methods showed that in this group it is possible to find both female and male heterogametic sex determining mechanisms. In *O. hornorum*, *O. aureus*, *O. karongae*, and *Tilapia mariae* the female is heterogametic with a WZ-ZZ system, whereas in *O. niloticus* and *T. zillii* the male is heterogametic following an XX-XY system (Mair et al., 1991a,b; Desprez et al., 2003; Cnaani et al., 2008).

First Foresti et al. (1993), using XY males, and subsequently Carrasco et al. (1999), with XX, XY and YY individuals, have tried to identify the X and Y chromosomes by synaptonemal complex analysis of meiotic chromosomes. These last authors found an absence of pairing in the terminal portion of the largest chromosome pair in 25% of the pachytene preparations obtained from XY males, while normal pairing was observed in homogametic individuals from both XX and YY genotypes. They suggested that the inhibition of pairing of this large chromosome was due to a small accumulation of heterochromatin and corresponded to the sex-determining region (Griffin et al., 2002). The large chromosome pair does in fact show an accumulation of retrotransposons and other repetitive sequences which is a common feature of sex chromosomes (Bezault et al., 2001; Martins et al., 2004; Cnaani et al., 2008).

Sex-linked markers have recently been identified in *O. niloticus* and *O. aureus* (Lee et al., 2003, 2004; Shirak et al., 2006; Cnaani et al., 2008). Segregation differences were found in a region of linkage group 1 (LG1) in *O. niloticus* (Lee et al., 2003). In 95% of the individuals from 2 out of 3 families studied, it was possible to predict the phenotypic sex from the marker information, but in the third family, markers did not segregate for the same Y-haplotype suggesting that additional sex-determining factors were acting. In *O. aureus* two unlinked loci were found to interact to determine sex (Lee et al., 2004). A major female determinant (W-haplotype) was found in LG3 which was epistatic to a dominant male determiner (Y-haplotype) in LG1. A contribution of LG23 to sex determination has also been found for two different QTLs (Cnaani et al., 2004) with the mapping of two genes of the vertebrate sex determination cascade (Shirak et al., 2006). Recently, Cnaani et al. (2008) have confirmed the complexity of sex determination by DNA marker segregation patterns showing variations in both the strain and the species amongst the tilapia group. They also physically mapped sex-linked markers using BACs with FISH, anchoring LG3 to the large sex chromosome pair and LG1 to a small pair.

In the past, numerous cases have indeed shown large deviations in sex ratios experimentally in Nile tilapia which could not be predicted by a simple mono-factorial model. Segregation of sex-linked DNA markers (Lee et al., 2003, 2004; Shirak et al., 2006; Cnaani et al., 2008) has confirmed what has been postulated previously on the existence of a single or a multi-allelic major sex determinant as well as an additional epistatic locus (or perhaps several loci) presumably autosomal (Hammerman and Avtalion, 1979; Mair et al., 1991a; Baroiller et al., 1995a, 1996; Mair et al., 1997; Baroiller and D'Cotta, 2001).

3. Temperature influences on sex ratios in domestic strains: windows of sexual lability and heritability

Despite the strong genetic basis for determining sex in tilapia it is now clear that other factors are also acting on sex. A strong effect of temperature on sex differentiation has been demonstrated in various tilapia species and in a hybrid (Baroiller et al., 1995a,b; 1996; Baroiller and Clota, 1998; Desprez and Mélard, 1998; Wang and Tsai, 2000). Following these results, influence of environmental factors on sex differentiation has been reported for more than 60 different teleost species (see reviews by Baroiller et al., 1999; Strüssmann and Patiño, 1999; Baroiller and D'Cotta, 2001; Godwin et al., 2003; Conover, 2004). Unfortunately most of these studies showing sensitivity to temperature are hindered because the sex determination mechanisms of most of these species have not been able to be well characterized (i.e. zebrafish *Danio rerio*, *Tetraodon*, ...) and physiological, genetic or ecological studies cannot be carried out to better understand the component of this environmental sensitivity.

In 1995, Baroiller et al., demonstrated that tilapias were sensitive to temperature during the critical period of sex differentiation. It was possible to masculinise XX progenies (100% females) with elevated temperatures above 32 °C, giving functional male phenotypes. The use of female monosex populations as well as the progeny testing of temperature treated males, has definitely demonstrated the existence of skewed male sex ratios corresponding to a sex-inversion of genetic females (XX) to functional phenotypic males. These thermo-neomales ($\Delta\sigma$ XX) provide in their offspring all-female or almost all-female progenies depending on the breeders (Baroiller et al., 1995a,b). High temperatures could efficiently masculinise some progenies if started around 10 days post fertilization (dpf) and if applied for at least 10 days, with longer periods being just as effective (Baroiller et al., 1995a,b; Tessema et al., 2006; Wessels and Hörstgen-Schwark, 2007). However, if a treatment was applied for a 10-day period but begun at 7 dpf, it had no effect on sex ratios (Baroiller et al., 1995a,b). This window for temperature sensitivity coincides with the gonad sensitivity towards other external factors notably hormones. Like temperature, hormonal treatments or the use of aromatase inhibitors during sex differentiation can override the genetic sex determination, inverting sex and producing functional phenotypes (Nakamura, 1975; Baroiller et al., 1999; Guiguen et al., 1999). Indeed, studies initiated by Yamamoto (1969) have shown that in fish, gonadal sex differentiation is extremely malleable and can be manipulated by hormones as well

Table 1

Sex ratios of temperature treated and untreated progenies from repeated matings from a domestic strain (Bouake) and a wild population of Nile tilapia.

Strain/population	Father	Mother	Control (27–28 °C)		36 °C treatment		Reference
	No.	No.	Nb	%	Nb	%	
				Males		Males	
Lake Manzala, Egypt	1	35	368	50.8	352	89.8 ^a	Tessema et al., 2006
	1	35	179	51.4	202	86.1 ^a	
	1	40	324	50.9	321	72.3 ^a	
	1	40	276	48.9	275	68.7 ^a	
	7	10	216	50	246	94.3 ^a	
Bouake, Ivory Coast	7	10	107	50.5	94	92.6 ^a	Baroiller and Clota, Unpub. Data ^b
	A	b	100	64	100	73 ^a	
	A	b	100	64	100	76 ^a	
	B	c	100	58	100	80 ^a	
	B	c	100	65	100	83 ^a	
	C	d	100	49	100	63	
	C	d	100	50	100	59	
	D	e	100	59	100	65	
	D	e	100	60	100	65	

Nb = number of sexed individuals.

^a significantly different from controls (χ^2 -test; $p < 0.05$).

^b Material and methods already published in Baroiller et al. (1995a); D'Cotta et al. (2001a) and Bezault et al. (2007).

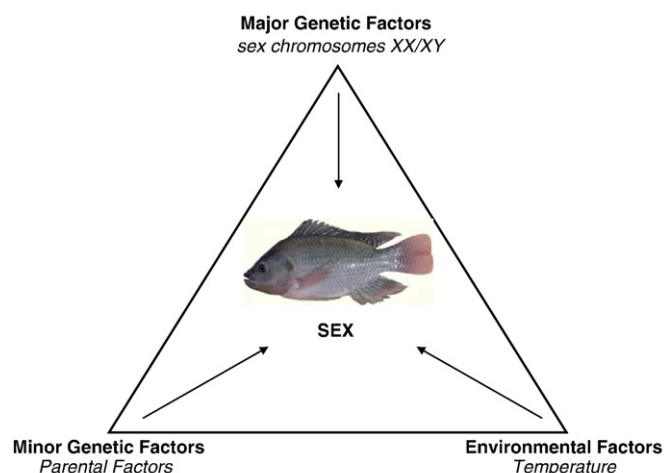


Fig. 1. Schematic triangle representing the complexity of tilapia sex determination showing the three factors influencing sex: the genetic sex determination carried by the XX/XY sex chromosomes, the minor genetic factors which are parental, and temperature, the environmental factor.

as other exogenous factors (Baroiller and D'Cotta, 2001; Devlin and Nagahama, 2002). The temperature sensitivity of Nile tilapia during sex differentiation is not seen in all progenies. Some male or female breeders provide progenies displaying a high sensitivity to temperature giving a high proportion of males in their sex ratio, while others gave an insensitive balanced sex ratio. A diallel crossing schema (5×5) of breeders demonstrated that the percentage of males was very different at the inter-individual level, indicating that there was an important parental effect (Baroiller and D'Cotta, 2001). This was confirmed with a comprehensive investigation on two populations of *O. niloticus* by Tessema et al. (2006). Both dam and sire contribute to these genetic parental effects (Table 1).

Wessels and Hörstgen-Schwark (2007) provided evidence that a surplus of males in temperature treated groups can be selected for, as a quantitative trait. They were the first to determine heritability for temperature sensitivity in Nile tilapia. A heritability of 0.69 was obtained through a two-generation selection experiment for high temperature sensitivity. Similar response to selection is seen in the third generation of selection for sex-ratio in the temperature-treated group (Wessels and Hörstgen-Schwark, unpublished data). Genetic correlations with other life history traits were not assessed when the selection experiment was conducted.

Together, these studies showed that sex in the Nile tilapia is governed by the interactions of 3 components, a complex genetic sex determination system with a major determinant locus and some minor genetic factors, as well as the influence of temperature (Fig. 1).

Extreme temperatures have also shown contrasting feminising effects using classic genotypes of *O. mossambicus* (Wang and Tsai,

2000), as well as in XY (Abucay et al., 1999; Wessels and Hörstgen-Schwark, 2008) and YY offsprings of *O. niloticus* (Kwon et al., 2000). In *O. mossambicus* a low temperature treatment of 5 days at 20 °C initiated before 14 dpf induced a feminisation, whereas high temperature treatments of 5 days at 32 °C applied after 14 dpf induced a masculinisation (Wang and Tsai, 2000). However, no feminisation has been observed in either *O. niloticus* or in *O. aureus*, nor in the red tilapia (Florida strain), with low temperatures ranging from 18 to 23 °C (Baroiller et al., 1995b; Desprez and Méléard, 1998; Abucay et al., 1999; Tessema et al., 2006).

Recently, Rougeot et al. (2008), demonstrated that even a precocious elevated temperature-treatment applied 12 h post fertilization (hpf) and kept for 52±2 h (till hatching), can also induce significantly skewed sex ratios towards males (8–27%, $n=4$) on a true all-female progeny (100% females in the control group). Even if the high mortality associated to the precociously high temperature treatment (35–36 °C) has to be considered (Table 2), this data strongly suggest that there is a thermosensitivity window very shortly after fertilization (between 12 and 52 hpf (= 4 dpf)), long before the development of the presumptive gonads.

At this stage (12 hpf), Nile tilapia larvae still have a brain rudiment (31 hpf) as well as primordial germ cells (46 hpf) (Morrison et al., 2001). Therefore, it can be considered that high temperatures can either precociously act on the developing brain or/and on the segregated primordial germ cells. Interestingly two similar windows are also identified for hormonal treatments (Rougeot et al., 2007).

These results raise three important questions regarding temperature effects: 1) Is temperature acting on sex determination, differentiation or both? 2) Are there different thermo-sensitive stages in the cascade of sex determination? 3) What is (are) the true target organ(s) (gonads, central nervous system, both organs) for temperature effects? We believe that the group of tilapias can be a good model to answer these questions.

However, similar precocious treatments with lower temperatures (34 °C) do not induce sex ratio deviations using progenies from both sensitive and non-sensitive breeders (Wessels, Samavati, Hörstgen-Schwark, unpublished data).

4. Evidence of temperature-sensitivity in wild populations

Most studies on tilapia sex determination or differentiation have been conducted on domestic or aquaculture strains. Several inter-generic hybridizations were performed worldwide to produce male offsprings (see Cnaani et al., 2008) and since hybrids were mostly fertile, it has probably favoured gene flow between the parental species. Some of these strains have been also used for selection programmes. Therefore, the original genetic diversity of these stocks might have been modified through inbreeding, genetic drift, introgression, hybridization, and/or selection. This could have affected the balance between the different factors (Fig. 1) involved in the sex

Table 2

Survival rates at the end of the temperature treatment, from a domestic strain (Bouake) and wild populations of Nile tilapia.

Strain/population	Treatment	Survival rate (mean)	Mini-maxi	Reference
Lake Manzala, Egypt	Control (28 °C)	95%	69–100%	Tessema et al., 2006
	18 °C for 20 days starting at 10 dpf	91%	67–99%	
	36 °C for 10 days starting at 10 dpf	95%	86–100%	
	38 °C for 10 days starting at 10 dpf	87%	74–100%	
Bouake, IvoryCoast	Control (27 °C)	96%	93–99%	Baroiller and Clota, Unpub. Data ^a
	36 °C for 30 days starting at 10 dpf	90%	87–95%	
Lake Volta, Ghana	Control (27 °C)	92%		Bezault et al., 2007
	36 °C for 30 days starting at 10 dpf	95%		
Lake Manzala, Egypt	Control (27 °C)	50%	16–72%	Rougeot et al., 2008
	34 °C till hatching, starting before 12h pf	53%	19–72%	
	34 °C till hatching, starting before 12h pf	42%*	21–66%	
	34 °C till hatching, starting before 12h pf	30%*	11–57%	

^a Material and Methods already published in Baroiller et al. (1995a); D'Cotta et al. (2001a) and Bezault et al. (2007). * $p<0.05$.

determination system (Baroiller and D'Cotta, 2001; Tessema et al., 2006; Cnaani et al., 2008), especially in a group where two opposite homogametic systems have been found. Therefore it was interesting to analyse the sex determination system (SDS) in wild populations of Nile tilapia.

In fact, other than the studies on *Menidia menidia* (Conover, 2004) very few have analysed the SDS on wild fish populations. Wild populations of *O. niloticus* have colonised a wide range of habitats across Africa due to their large adaptive potential (Fig. 2), (Philippart and Ruwet, 1982; Trewavas, 1983; Baroiller and Toguyeni, 2004). They are found in habitats with: a) strong seasonal variations, e.g. Mediterranean in North of Egypt or sub-tropical regimes in West-Africa with alternation between hot (28–34 °C) and cold (22–26 °C) seasons, b) high altitude lakes with constant cold temperatures (17–24 °C), and c) hydrothermal hot springs (≥ 40 °C) (Trewavas, 1983; Bezault et al., 2007). Based on laboratory studies, some of these natural environmental conditions could very likely affect the sex ratios of these wild populations. They could indeed be exposed to masculinising temperatures during their thermosensitive period (10 to 20 dpf) considering tilapia reproductive behaviour and early life stages. Fry pass through a period of strict maternal mouth-brooding (from 0 dpf to 9–10 dpf) after which they shoal in shallow margins of water-bodies (Bruton and Bolt, 1975) encountering elevated (32–34 °C) temperatures potentially masculinising (Baroiller et al., 1995a,b; Bezault, 2005). In fact the fry will experience a fluctuating thermal regime according to nycthemeral temperature variations and micro-habitat migrations (Bruton and Bolt, 1975) which contrasts with the constant temperatures applied during laboratory experimentations. Nevertheless, this type of fluctuant thermal regime (i.e. 35 °C, during the day and 27 °C overnight) was shown to exert

significant effects on the sex ratio in the blue tilapia, *O. aureus* (Baras et al., 2000).

Temperature effects on sex differentiation have been analysed in six wild populations of Nile tilapia adapted to different thermal conditions (Fig. 2) (Altena and Hörstgen-Schwark, 2002; Tessema et al., 2006; Bezault et al., 2007). Four wild populations live under important and very distinct seasonal variations: Lake Manzala (Egypt), Lake Rudolph (Kenya) at the Southern end of the Nilotic region; Lake Victoria (Winiam Gulf, Kenya) in the East-African Rift Valley and Lake Volta (Kpandu, Ghana) in West Africa. The two other populations are located in the Ethiopian highlands with constant but alternative extreme temperature conditions: Lake Koka, a high altitude cold-water lake, and Lake Metahara, exclusively fed by the resurgence of hydrothermal hot springs (43 °C) (Bezault, 2005). Using tagged breeders, sex ratios were analysed from progenies reared at control (27–28 °C) and high-temperatures (36 °C) applied for 10 or 30 day periods, resulting in the same masculinisation rate (Table 3) (Baroiller et al., 1995a,b; Altena and Hörstgen-Schwark, 2002; Tessema et al., 2006). In the controls, all the wild populations showed a relatively balanced sex ratio (1:1) or slightly skewed towards males except the Koka, Metahara and Kpandu populations where some progenies had highly skewed sex-ratios (Bezault et al., 2007), which could be attributed to parental effects. The inter-population crossings support the hypothesis of a predominant monofactorial sex determining factor shared among all *O. niloticus* populations, without ruling out the implication of additional minor genetic factors in the SDS (Tessema et al., 2006; Bezault et al., 2007). Under high-temperature treatments, increase of male proportions was observed in all the populations compared to their controls (Table 3). A higher mortality was only

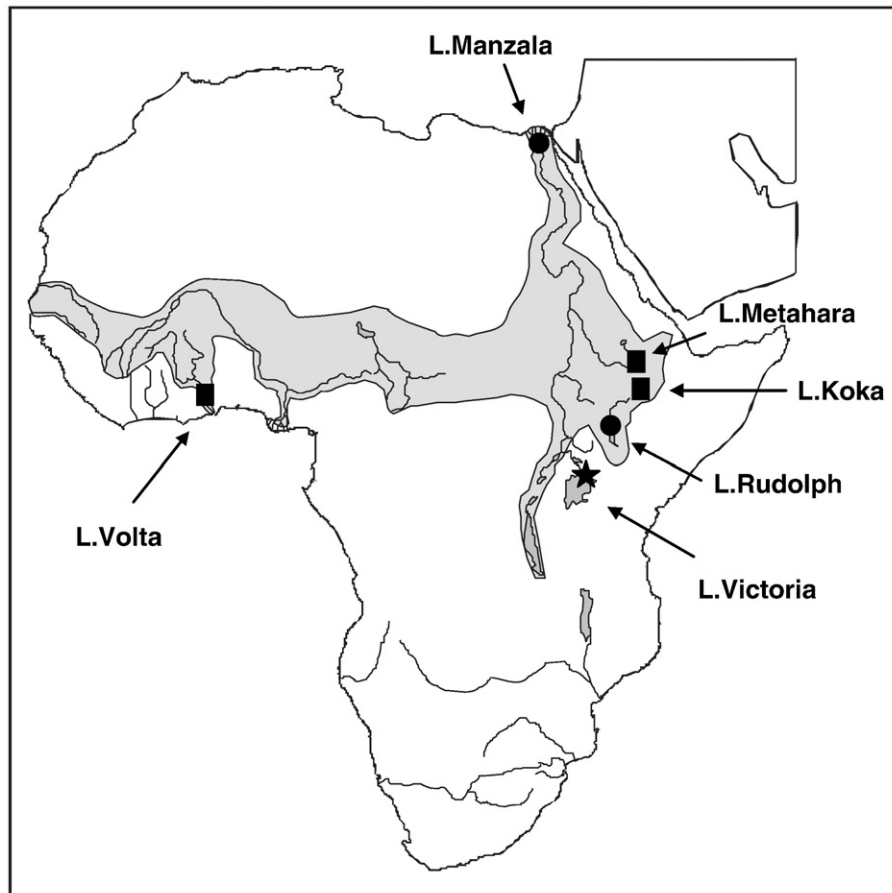


Fig. 2. Geographic distribution of the Nile tilapia, *Oreochromis niloticus* (in grey), with the sampling locations of the six wild populations investigated to test their sex determination: Lakes Manzala (Egypt), Rudolph (Kenya), Victoria (Kenya), Volta (Ghana), Koka (Ethiopia) and Metahara (Ethiopia). Symbols correspond to References ★ Altena and Hörstgen-Schwark (2002); ● Tessema et al. (2006); and ■ Bezault et al. (2007).

Table 3
Effect of the temperature treatment on progeny sex ratios from 6 wild populations of Nile tilapia.

Populations		Thermic regime	Nb progeny tested	Control (27–28 °C)		Temperature treatment (36 °C)				Reference
Location	Drainage			Sex ratio	% Progeny where	Sex ratio	% Progeny where			
				% Males	SR ≠ (1:1)	% Males	SR _T >SR _C	SR _T >80%	SR _T >95%	
Koka	Awash	Cold	18	55.8%	22%	77%*	67%	50%	22%	Bezault et al., 2007
Metahara	Awash	Hot	18	52.9%	27%	80.6%*	83%	67%	0%	
Kpandu	Volta	Variable	21	50.2%	19%	78.7%*	71%	48%	19%	Tessema et al., 2006
Manzala	Nil	Variable	15	49.9%	0%	78.4%*	87%	67%	20%	
Rudolph	Rudolph	Variable	11	53.0%	0%	61.4%*	27%	0%	0%	Altena and Hörstgen-Schwark, 2002
Victoria	Victoria	Variable	11	54.1%	–	78.4%*	55%	18%	–	

* *p* < 0.05.

found for the treated groups from the cold-water population of Lac Koka (Bezault et al., 2007). For all the other populations, we could clearly rule out the hypothesis of a differential mortality between sexes to explain the increased proportion of males at high temperatures. This is similar to what has been demonstrated (Table 2) when using either genetically all-female progenies (Baroiller et al., 1996) or by identifying several XX males in the temperature treated groups (Baroiller et al., 1995a,b; Tessema et al., 2006). These results demonstrate the existence of both sensitive and insensitive progenies in all the tested wild populations. Important differences between the populations were found concerning the proportion of thermosensitive progenies (only 27% in Lake Rudolph against 94% in the Metahara population) or the proportion of males in these progenies (Table 3). On average, the sex ratio of the treated groups varied between 61.4% and 80.6% (Table 3). These differences can either be related to their respective proportion of thermosensitive individuals or/and to their temperature threshold (intensity or duration) for sex inversion, as already shown in domestic species or strains (e.g. 34°–35 °C for *O. aureus*, and for *O. niloticus*, 36 °C for the Bouaké strain versus 37 °C for the Manzala one) (Baroiller et al., 1995a,b; Abucay et al., 1999; Baras et al., 2001; Baras et al., 2002). This can reflect local adaptations and variability in the Genetic-by-Environment (G×E) interactions.

A clear parental effect on thermosensitivity with an influence of both parents has been demonstrated at the level of individuals, based on half-sibs progenies (Baroiller and D'Cotta, 2001; Tessema et al., 2006; Bezault et al., 2007), and on repeated crosses from given mating partners (Tessema et al., 2006) (Tables 1 and 4).

Some wild breeders collected in Lake Volta and Lake Koka populations gave progenies with a highly skewed sex ratio at 27 °C (Bezault et al., 2007), strongly suggesting the existence of spontaneously sex-inversed individuals, with at least one neo-male (Δ♂ XX) and one neo-female (Δ♀ XY) (Bezault et al., 2007). Unfortunately it was impossible to definitely conclude whether these sex reversals were the result of minor genetic factors or a temperature influence.

Table 4
Parental influences on high temperature effects in a domestic strain (Bouake) and wild populations of Nile tilapia.

Strain/population	Father	Mother	Control (27–28 °C) % Males	36 °C Treatment % Males	Reference
	No.	No.			
Lake Koka Ethiopia	5	12	55	58	Bezault et al., 2007
	5	4	55	84*	
	39	4	72	98*	
	39	6	67	100*	
Lake Manzala, Egypt	3	1	49.3	55.6	Tessema et al., 2006
	3	5	48.6	45.3	
	4	1	50.8	61.4*	
	4	2	48.9	94*	
	10	1	47.7	91.3*	
Bouake, Ivory Coast	a	B	70	82*	Baroiller and D'Cotta, 2001
	a	C	68	73	
	c	B	67	98*	
	c	C	56	85*	

* Significantly different from controls (χ²-test; *p* < 0.05).

These different sex determining studies conducted on wild populations of Nile tilapia emphasize the existence of a complex SDS, combining GSD and thermosensitivity. This is based on the strong assumption that individuals are likely, through geographic or seasonal conditions, to experience temperatures able to influence sex ratios. The next step will be to evaluate how frequently temperatures can override the genetic sex determination in wild populations.

5. Temperature action on the sex determining/differentiating cascade in tilapia

The trigger of sex determination in tilapias is not known. As mentioned previously the major genetic locus in Nile tilapia is present on LG1 but several other loci (LG3, LG23) are probably acting simultaneously to determine sex. Despite the fact that the fate of the gonad is determined genetically, temperature can override it and switch the mechanism when the gonad is undifferentiated. But once the “decision” is established it cannot be modified anymore, being committed towards the development of one sex. The really critical period of gonad differentiation in tilapia has been established from 9 to 15 dpf (D'Cotta et al., 2007; Ijiri et al., 2008). Temperature or hormonal treatments have to be applied from this period onwards to be efficient. This is just before the appearance of the very first sex-specific difference, an active mitosis in the ovary (Nakamura et al., 1998; D'Cotta et al., 2001a; Ijiri et al., 2008).

Although sex can be determined by multiple mechanisms in vertebrates, the gonads are structurally and functionally very similar. Therefore, it was postulated that the underlying developmental mechanisms downstream of the sex determinant, were probably similar. Many genes involved in the sex-determining cascade have now been identified and characterized in mammals. Through comparative studies it has been possible to clone orthologue genes in non-mammalian vertebrates, including teleosts. Despite a high conservation of expression patterns during gonad development, spatial and temporal differences suggest that some of them have different roles and regulation.

It has long been known that gonad estrogens acted as natural inducers for ovarian development in lower vertebrates (Yamamoto, 1969; Nakamura et al., 1998). The aromatase enzyme (= *Cyp19* gene) catalyzes the conversion of androgens into 17β-estradiol (Baroiller et al., 1999) and if inhibited, blocks estrogen production causing a female to male sex reversal (Guiguen et al., 1999; Kwon et al., 2000). In developing ovaries of XX fish the *Cyp19a* (= *Cyp19a1a*) gene is up-regulated (D'Cotta et al., 2001a; Kwon et al., 2001; Ijiri et al., 2008). Ijiri et al. (2008) found higher levels of *Cyp19a* in future ovaries as early as 9 dpf and they increased rapidly until 19 dpf. Temperature applied during the sex differentiating period in tilapia XX offspring induced a down-regulation of *Cyp19a* (D'Cotta et al., 2001a) seen at 17 dpf (Baroiller et al., 2008). Furthermore, *Cyp19a* expression levels were correlated with the proportion of temperature masculinised (TM) XX individuals (D'Cotta et al., 2008). Although two *Cyp19* genes have been found, only the ovarian form *Cyp19a* was dimorphic during ovarian sex differentiation (Kwon et al., 2001; Chang et al., 2005). *Cyp19a* promoter has binding regions for SF-1/Ad4 BP, WT1-KTS and SRY, which are sex determining

factors in mammals. A testis-determining factor could bind this *Cyp19a* promoter suppressing the gene and thereby, diminish 17 β -estradiol levels and drive male differentiation (Chang et al., 2005). A good candidate was *Wt1b* which encodes a zinc-finger DNA-binding protein involved in testis development and shown to up-regulate *SRY* expression by binding to DNA (Miyamoto et al., 2008). *Wt1b*, like *Cyp19a*, is in fact located on LG1 of Nile tilapia at close proximity to the major sex determining locus but, recombination studies excluded *Wt1b* as the sex determinant gene (Lee and Kocher, 2007).

An early player in the ovarian determining and differentiating pathway is *FoxL2*, a forkhead transcriptional factor involved in ovarian development and function in several vertebrates. It has an ovarian-specific expression in mammals, chickens and rainbow trout (Löffler et al., 2003; Baron et al., 2004). In “TSD turtles”, a dimorphic expression was seen in gonads at female promoting temperatures (Löffler et al., 2003). In tilapia, *FoxL2* is already expressed at 9 dpf in XX gonads only slightly higher than in XY but levels increased linearly hereafter in XX ovaries (Ijiri et al., 2008). Like in other vertebrates, *FoxL2* expression patterns were highly correlated with the *Cyp19a* expression (Ijiri et al., 2008; Baroiller et al., 2008). In TM XX tilapia *FoxL2* expression remained low like in XY gonads (Baroiller et al., 2008). *In vitro* studies have demonstrated that *FoxL2*, binds to *Cyp19a* promoter and activates its transcription (Wang et al., 2007). (Fig. 3).

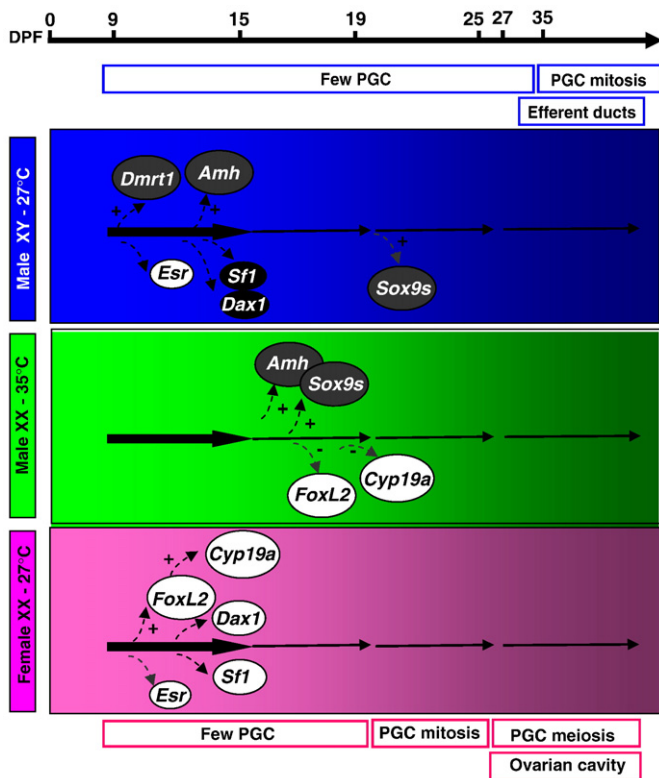


Fig. 3. Schematic pattern of gene expressions found in the tilapia gonads of XX females (pink rectangle), XY males (blue rectangle) and temperature-masculinised XX fish (green rectangle), during the sex determining/differentiating pathway. Data is a compilation from D'Cotta et al. (2007); Ijiri et al. (2008); and Baroiller et al. (2008). The large arrow corresponds to what is considered the critical period of sex differentiation. The top scale represents days post-fertilization (dpf). The white circles are considered ovarian developmental genes while black circles are considered testis developmental genes. + sign means an up-regulation from then onwards while – sign means down-regulation from then onwards. All other genes were expressed at similar levels after their appearance. Sox9 profiles are a compilation of data from all Sox9 forms. *Amh* anti-Müllerian hormone gene; *Cyp19a* (= *Cyp19a1*) ovarian aromatase form; *FoxL2* forkhead transcriptional factor L2; *Dmrt1* Doublesex mab3 related transcription factor 1; *Sox9* Sry-related HMG-box protein 9 gene; *Dax1* dosage-sensitive sex reversal, adrenal hypoplasia congenital critical region on the X chromosome; *Sf1* steroidogenic factor 1 = *nr5a1*; *Esr* Estrogen receptors; PGC primordial germ cells.

The Sry-related HMG-box protein 9 gene, *SOX9* plays a role in the male cascade of vertebrates. In mammals, *Sox9* expression is seen immediately after that of *Sry* and may be a downstream effector; in mice *Sox9* mediates the beginning of *Amh* expression in Sertoli cells (Yao and Capel, 2005). In tilapia XX and XY gonads, *Sox9* expression levels were similar from 9 to 29 dpf, becoming stronger thereafter in XY males (Ijiri et al., 2008). In contrast, we found higher levels of *Sox9a* and *Sox9b* expressions in XY gonads from 20 to 25 dpf (D'Cotta et al., 2007). Differences between both studies may derive from the use of form-specific primers or from the strains. In TM XX individuals, increase was already evident at 17 dpf for both *Sox9s*.

Another actor of the male differentiating pathway is the anti-Müllerian hormone gene, *Amh* responsible for the regression of the Mullerian ducts in males. *Amh* was found in fish which do not have Mullerian ducts, and its role is still unknown. In TM XX gonads both *Amh* and *Sox9s* increased later than in XY males. *Amh* increase in XY male gonads was already evident from 10 to 15 dpf (D'Cotta et al., 2007), or after 19 dpf (Ijiri et al., 2008). Taken together, these results show that in tilapia *Amh* increases in XY male gonads before that of either *Sox9a* or *Sox9b*, similar to reports on chicken, and on “TSD reptiles”, i.e. alligator and red-eared turtle (Smith and Sinclair, 2004; Western et al., 1999; Shoemaker et al., 2007). *Sox9* is suggested to have a role in testicular tubules formation rather than male determination or differentiation (Ijiri et al., 2008). In contrast, since *Amh* is up-regulated in supporting cells it probably has a role in testicular differentiation (Ijiri et al., 2008). Interestingly, *Amh* has been mapped to LG23 where two QTLs for sex exists (Shirak et al., 2006).

Between 8 and 26 dpf, elevated levels of 11ketotestosterone (11KT, an important androgen in tilapia) were measured in XY males (Baroiller and D'Cotta, 2001) but they are unlikely to drive male differentiation. This is because no expression was seen in XX, XY or XX TM of 11 β -hydroxylase (*Cyp11b2*) responsible for 11 β -hydroxytestosterone synthesis (a precursor of 11KT) at the critical period. It appears only at 39 dpf at onset of testis mitosis (D'Cotta et al., 2001b; Ijiri et al., 2008).

Although the Doublesex mab3 related transcription factor 1 (*Dmrt1*) has not been analysed yet in TM XX tilapia, Ijiri et al. (2008) found an early male-specific expression in XY tilapia and considered it as one of the critical players in male differentiation. *Dmrt1* is closely related to the male determinant *DMY/Dmrt1bY* of medaka (Matsuda et al., 2002; Nanda et al., 2002) and has a male-specific expression at early stages in chicken and turtles during testes development, but it does not play a role in mammalian testis determination (Yao and Capel, 2005; Shoemaker et al., 2007). In “TSD turtles” it is rapidly up-regulated at male promoting temperatures (Shoemaker et al., 2007).

Based on the findings that brain aromatase was repressed in TM XX tilapia (D'Cotta et al., 2001a), expression profiles were studied thereafter simultaneously in the brain and gonad, and to date all the sex differentiating genes of the gonad were also present in the brain (D'Cotta et al., 2007; Baroiller et al., 2008). Furthermore, *Sox9b*, *Amh* and *Dax1* showed strong dimorphic high expression in the brain of XY fish much earlier than in the gonad, around 10 to 15 dpf. No sex differences were seen after this period. XX and TM-XX brains showed no expression differences probably due to temperature treatments being applied at 10 dpf, a period probably too short to elicit an effect at 15 dpf.

6. Coexistence of GSD and TSD in tilapia?

In reptiles, the discovery that sex ratios can be determined by precocious exposure of eggs or embryos to either high or low temperatures is about 40 years old and was found in a lizard (Charnier, 1966) and in a turtle (Pieau, 1971, 1972). Almost simultaneously Ohno (1967) reported the existence of heteromorphic sex chromosomes in several other reptiles (various snakes, lizards and a few turtles). Under natural conditions, discordance between the sexual genotype and the gonadal phenotype (XX male, XY female, ZW male or ZZ female) have never been found, suggesting that reptiles had either a genetic sex

determination (GSD) with sex chromosomes or a temperature sex determination (TSD). Both TSD and GSD systems were considered to be separated, but a continuum between them with intermediate mechanisms may occur (Bull, 1980; Sarre et al., 2004; Valenzuela, 2004). Until recently, transitional forms had never been reported (Shine et al., 2002; Sarre et al., 2004). However, as stated very recently by Bull (2008) one of the fathers of the adaptive hypothesis of TSD (Charnov and Bull, 1977), empirical evidences against coexistence of TSD and GSD were very limited. This was due to the low number of samples analysed under natural conditions and because sex had to be identified through invasive methods in juveniles which do not exhibit sexual dimorphisms (Bull, 2008). Therefore, the importance of possible discordance between the sexual genotype and gonadal phenotype is not known in reptiles (Bull, 2008). However, recent studies in two distantly related lizard taxa with two opposite SDS (XX/XY and ZZ/ZW) have demonstrated that extreme temperatures can override the GSD (Shine et al., 2002; Sarre et al., 2004; Quinn et al., 2007; Radder et al., 2008). In the three-lined skink, *Bassiana duperreyi*, sex is generally determined by the presence or absence of the Y chromosome (male heterogamety) and balanced sex ratios are observed in most natural nests. However, cool temperature regimes induced male skewed sex ratios (Shine et al., 2002). Sex-specific DNA markers confirmed the sex inversion of some XX individuals by low temperatures (Radder et al., 2008). Another case is the dragon lizard, *Pogona vitticeps* where a female heterogamety (ZW/ZZ) was found (Ezaz et al., 2005) but high temperatures induced female skewed sex ratios and W-specific DNA markers confirmed sex inversion of ZZ individuals by high temperatures (Quinn et al., 2007). Since incubation treatments of *B. duperreyi* mimic the thermal regimes found in natural nests (Radder et al., 2008), it is possible that masculinisation by low temperatures could exist occasionally in nature (Bull, 2008). These temperature effects in species with sex chromosomes challenges the theory of a strict dichotomy between the two sex determination systems and supports the hypothesis of a continuum between them with possible coexistence of TSD and GSD in reptiles.

In fish, TSD has been first reported by Conover and Kynard (1981) on Atlantic silverside, *Menidia menidia*, based on the classic dichotomic classification for sex-determining systems in reptiles. Surprisingly, following this major paper and the subsequent work done on this species (Conover, 2004), almost 15 years passed before the publication of a thermal influence in another fish species, the tilapia (Baroiller et al., 1995a,b) and subsequently in the pejerrey, *Odontesthes bonariensis* (Strüssmann et al., 1996). These two papers suggested that a temperature influence on sex ratio could be more widespread than expected and stimulated various studies on more than 60 species, for either basic or applied research (Baroiller and D'Cotta, 2001; Ospina-Alvarez and Piferrer, 2008). Besides Atlantic silverside, tilapias and pejerrey, the Japanese hirame, *Paralichthys olivaceus*, and the European sea bass, *Dicentrarchus labrax*, have also become major models to study the mechanisms of thermal influences on sex ratios. For these species knowledge is still scarce, compared to the 40 years of study on reptiles. Together the studies on tilapias sex determination suggest that it resembles that of the three-lined skink and the bearded dragon lizard, where sex ratios are governed by sex chromosomes (either a male or a female homogamety) and modified by extreme temperatures encountered by these species at least in some natural conditions. Although in tilapia additional minor genetic factors are clearly acting also on sex.

In tilapia like in various reptiles (Valenzuela, 2008), genetic x environment interactions have been demonstrated in the sex-determining response to temperature modulation. In all of these species, treatments that mimic the natural thermal regimes have demonstrated to be efficient for sex inversion. Furthermore, the recent data of Bezault et al. (2007) strongly suggest that in tilapias XX males and XY females can be encountered in nature. As XX males and YY males have clearly been demonstrated to be viable and fertile in tilapias, the existence of XX males and YY males in the wild is not a

drawback for the population and species. Bull (2008) suggests that in the two lizards, sex determination can be controlled by TSD at extreme temperatures (and be adaptive at least in lizards) as well as by sex chromosomes in the middle temperature ranges. We believe a very similar complex sex determination system can exist in tilapias, also supporting the hypothesis of a continuum between TSD and GSD.

As stated by Bull (2008), whether the coexistence of TSD and GSD is an "accident" (infrequent relict) or an adaptation remains the major issue (Warner and Shine, 2008). To explain the coexistence of sex chromosomes and TSD, benefits of both systems have to be combined (Bull, 2008). In tilapia sex chromosomes carry useful (thus valuable) genes for males on the Y (i.e. for growth: Toguyeni et al., 2002) which can favour their retention. We still have to understand, why at extreme temperatures an XX individual would have a better fitness as a male rather than as a female. This will explain why temperature effects are conserved with the genetic sex determination. We believe that the group of tilapias constitutes an excellent model to answer these questions and to better understand the mechanisms of sex differentiation and sex determination under classic or temperature-induced conditions. The tilapia genome sequence which will be available in 2009 and the current genetics maps will contribute substantially to understand how temperature and genetics meet.

Acknowledgements

Supported by the German Research Foundation grants to GHS (Ho 838/5) and the French ANR-Genanimal, Project Fishsex (ANR-06-GANI-012).

References

- Abucay, J.S., Mair, G.C., Skibinski, D.O., Beardmore, J.A., 1999. Environmental sex determination: the effect of temperature and salinity on sex ratio in *Oreochromis niloticus* L. *Aquaculture* 173, 219–234.
- Altena, A., Hörstgen-Schwark, G., 2002. Effects of rearing temperatures on sex ratios in tilapia, *Oreochromis niloticus* L., investigations on a local population from the Lake Victoria in Kenya. Challenge to Organic Farming and Sustainable Land Use in the Tropics and Subtropics. Deutscher Tropentag, Witzhausen.
- Baras, E., Prignon, C., Gohoungou, G., Méléard, C., 2000. Phenotypic sex differentiation of blue tilapia under constant and fluctuating thermal regimes and its adaptive and evolutionary implications. *J. Fish Biol.* 57, 210–223.
- Baras, E., Jacobs, B., Méléard, C., 2001. Effect of water temperature on survival, growth and phenotypic sex of mixed (XX-XY) progenies of Nile tilapia *Oreochromis niloticus*. *Aquaculture* 192, 187–199.
- Baras, E., Mpo'n'tcha, A., Driouch, H., Prignon, C., Méléard, C., 2002. Ontogenetic variations of thermal optimum for growth, and its implication on thermolabile sex determination in blue tilapia. *J. Fish Biol.* 61, 645–660.
- Baroiller, J.F., Clota, F., 1998. Interactions between temperature effects and genotype on *Oreochromis niloticus* sex determination. *J. Exp. Zool.* 281, 507.
- Baroiller, J.F., D'Cotta, H., 2001. Environment and sex determination in farmed fish. *Comp. Biochem. Physiol. C: Toxicol. Pharmacol.* 130, 399–409.
- Baroiller, J.F., Jalabert, B., 1989. Contribution of research in reproductive physiology to the culture of tilapias. *Aquat. Living Resour.* 2, 105–116.
- Baroiller, J.F., Toguyeni, A., 2004. The Tilapiini tribe: environmental and social aspects of reproduction and growth. In: Safran, P. (Ed.), *Fisheries and Aquaculture, Encyclopedia of Life Support Systems (EOLSS)*. Developed under the Auspices of the UNESCO. Eolss Publishers, Oxford, UK. [<http://www.eolss.net>].
- Baroiller, J.F., Chourrout, D., Fostier, A., Jalabert, B., 1995a. Temperature and sex chromosomes govern sex ratios of the mouthbrooding cichlid fish *Oreochromis niloticus*. *J. Exp. Zool.* 273, 216–223.
- Baroiller, J.F., Clota, F., Geraz, E., 1995b. Temperature sex determination in two tilapia species, *Oreochromis niloticus* and the red tilapia (Red Florida strain): effect of high or low temperatures. In: Goetz, F.W., Thomas, P. (Eds.), *Proceedings of the 5th International Symposium on the Reproductive Physiology of Fish*. The University of Texas, Austin, Texas, USA, pp. 158–160. July 2–8, 1995.
- Baroiller, J.F., Nakayama, I., Foresti, F., Chourrout, D., 1996. Sex determination studies in two species of teleost fish, *Oreochromis niloticus* and *Leporinus elongatus*. *Zool. Stud.* 35, 279–285.
- Baroiller, J.F., Guiguen, Y., Fostier, A., 1999. Endocrine and environmental aspects of sex differentiation in fish. *Cell. Mol. Life Sci.* 55, 910–931.
- Baroiller, J.F., D'Cotta, H., Wessels, S., Bezault, E., Hörstgen-Schwark, G., 2008. Temperature effects on tilapia sex differentiation. *Proceeding of the 6th International Symposium on Fish Endocrinology*. Calgary, Canada, June 22–27.
- Baron, D., Cocquet, J., Xia, X., Fellous, M., Guiguen, Y., Veitia, R.A., 2004. An evolutionary and functional analysis of FoxL2 in rainbow trout gonad differentiation. *J. Mol. Endocrinol.* 33, 705–715.

- Beardmore, J.A., Mair, G.C., Lewis, R.I., 2001. Monosex male production in finfish as exemplified by tilapia: applications, problems, and prospects. *Aquaculture* 197, 283–301.
- Bezault, E., 2005. Etude du système de déterminisme du sexe au sein de populations naturelles de tilapia du Nil, *Oreochromis niloticus*: Importance des composantes génétiques et environnementales. Université Paris XI, Orsay, France, p. 270.
- Bezault, E., Ozouf-Costaz, C., D'Hont, A., Volff, J.N., Rognon, X., Baroiller, J.F., 2001. Structure and evolution of pure and hybrid genomes of Tilapia. *Chromosom. Res.* 9 (suppl. 1), 30.
- Bezault, E., Clota, F., Derivaz, M., Chevassus, B., Baroiller, J.F., 2007. Sex determination and temperature-induced sex differentiation in three natural populations of Nile tilapia (*Oreochromis niloticus*) adapted to extreme temperature conditions. *Aquaculture* 272, S3–S16.
- Bruton, M.N., Bolt, R.E., 1975. Aspects of the biology of *Tilapia mossambica* Peters (Pisces: Cichlidae) in a natural freshwater lake (Lake Sibaya, South Africa). *J. Fish. Biol.* 7, 423–445.
- Bull, J.J., 1980. Sex determination in reptiles. *Q. Rev. Biol.* 55, 3–21.
- Bull, J.J., 2008. Sex determination: are two mechanisms better than one? *J. Biosci.* 33, 5–8.
- Carrasco, L.A.P., Penman, D.J., Bromage, N.R., 1999. Evidence for the presence of sex chromosomes in the Nile tilapia (*Oreochromis niloticus*) from synaptonemal complex analysis of XX, XY and YY genotypes. *Aquaculture* 173, 207–218.
- Chang, X., Kobayashi, T., Senthilkumaran, B., Kobayashi-Kajura, H., Sudhakumari, C.C., Nagahama, Y., 2005. Two types of aromatase with different encoding genes, tissue distribution and developmental expression in Nile tilapia (*Oreochromis niloticus*). *Gen. Comp. Endocrinol.* 141, 101–115.
- Charnier, M., 1966. Action de la température sur la sex-ratio chez l'embryon d'Agama agama (Agamidae, Lacertiliens). *Soc. Biol. Quest. Af.* 160, 620–622.
- Charnov, E.L., Bull, J.J., 1977. When is sex environmentally determined? *Nature* 266, 828–830.
- Cnaani, A., Zilberman, N., Tinman, S., Hulata, G., Ron, M., 2004. Genome-scan analysis for quantitative trait loci in an F2 tilapia hybrid. *Mol. Genet. Genomics* 272, 162–172.
- Cnaani, A., Lee, B.Y., Zilberman, N., Ozouf-Costaz, C., Hulata, G., Ron, M., D'Hont, A., Baroiller, J.F., D'Cotta, H., Penman, D.J., Tomasino, E., Coutanceau, J.P., Pepey, E., Shirak, A., Kocher, T.D., 2008. Genetics of sex determination in tilapia species. *Sex. Dev.* 2, 43–54.
- Conover, D.O., 2004. Temperature-dependent sex determination in fishes. In: Valenzuela, N., Lance, V. (Eds.), *Temperature-Dependent Sex Determination in Vertebrates*. Smithsonian Books, Washington, DC, pp. 11–20.
- Conover, D.O., Kynard, B.E., 1981. Environmental sex determination—interaction of temperature and genotype in a fish. *Science* 213, 577–579.
- D'Cotta, H., Guiguen, Y., Govoroun, M.S., McMeel, O., Baroiller, J.F., 2001a. Aromatase plays a key role during normal and temperature-induced sex differentiation of tilapia *Oreochromis niloticus*. *Mol. Reprod. Dev.* 59, 265–276.
- D'Cotta, H., Fostier, A., Guiguen, Y., Govoroun, M., Baroiller, J.F., 2001b. Search for the molecular mechanisms implicated in the thermosensitivity of sex gonadal differentiation of tilapia fish: differential approach and expression of candidate genes. *J. Exp. Zool.* 290, 574–585.
- D'Cotta, H., Pepey, E., Pfennig, F., Bienvenu, D., Gutzeit, H.O., Volff, J.N., Wenning, M., Baroiller, J.F., 2007. *Sox9a*, *Sox9b* and *Amh* are up-regulated in the gonads during natural and temperature-induced tilapia male differentiation. *Proceedings of the 8th International Symposium on Reproductive Physiology of Fish*, June 3rd–8th, Saint-Malo, France.
- D'Cotta, H., Pepey, E., Wessels, S., Poonlaphdech, S., Reinelt, B., Hoerstgen-Schwark, G., Baroiller, J.F., 2008. Temperature-induced male differentiation in the Nile tilapia: gonad gene expression using female monosex populations and divergent thermosensitive lines. *Proceedings of the 20th International Congress of Zoology*, S10 Sex determination & differentiation, Paris, August 26–29.
- Desprez, D., Mèlard, C., 1998. Effect of ambient water temperature on sex determination in the blue tilapia, *Oreochromis aureus*. *Aquaculture* 162, 79–84.
- Desprez, D., Mèlard, C., Hoareau MC Bellemène, Y., Bosc, P., Baroiller, J.F., 2003. Inheritance of sex in two ZZ pseudofemale lines of tilapia *Oreochromis aureus*. *Aquaculture* 218, 131–140.
- Devlin, R.H., Nagahama, Y., 2002. Sex determination and sex differentiation in fish: an overview of genetic, physiological, and environmental influences. *Aquaculture* 208, 191–366.
- Ezaz, T., Quinn, A.E., Miura, I., Sarre, S.D., Georges, A., Marshall Graves, J.A., 2005. The dragon lizard *Pogona vitticeps* has ZZ/ZW micro-sex chromosomes. *Chromosom. Res.* 13, 763–776.
- FAO Fisheries and Aquaculture Information and Statistics Service [FAO FIES], 2008.
- Foresti, F., Oliveira, C., Galetti Jr, P.M., Almeida Toledo, L.F., 1993. Synaptonemal complex analysis in spermatocytes of tilapia, *Oreochromis niloticus* (Pisces, Cichlidae). *Genome* 36, 1124–1128 (che G10/P5).
- Godwin, J., Luckenbach, J.A., Borski, R.J., 2003. Ecology meets endocrinology: environmental sex determination in fishes. *Evol. Dev.* 5, 40–49.
- Griffin, D.K., Harvey, S.C., Campos-Ramos, R., Ayling, L.J., Bromage, N.R., Masabanda, J.S., Penman, D.J., 2002. Early origins of the X and Y chromosomes: lessons from tilapia. *Cytogenet. Genome Res.* 99, 157–163.
- Guiguen, Y., Baroiller, J.F., Ricordel, M.J., Iseki, K., Mcmeel, O.M., Martin, S.A., Fostier, A., 1999. Involvement of estrogens in the process of sex differentiation in two fish species: the rainbow trout (*Oncorhynchus mykiss*) and a tilapia (*Oreochromis niloticus*). *Mol. Reprod. Dev.* 54, 154–162.
- Hammerman, I.S., Avtalion, R.R., 1979. Sex determination in *Sarotherodon* (Tilapia). II. The sex ratio as a tool for the determination of genotype. A mathematical model of autosomal and gonosomal influence. *Theor. Appl. Genet.* 55, 177–187.
- Ijiri, S., Kaneko, H., Kobayashi, T., Wang, D.S., Sakai, F., Paul-Prasanth, B., Nakamura, M., Nagahama, Y., 2008. Sexual dimorphic expression of genes in gonads during early differentiation of a teleost fish, the Nile tilapia *Oreochromis niloticus*. *Biol. Reprod.* 78, 333–341.
- Jalabert, B., Moreau, J., Planquette, P., Billard, R., 1974. Déterminisme du sexe chez Tilapia macrochir et Tilapia nilotica: action de la méthyltestostérone dans l'alimentation des alevins sur la différenciation sexuelle; proportion des sexes dans la descendance des mâles "inversés". *Ann. Biol. Anim. Biochim. Biophys.* 14 (4-B), 729–739.
- Kwon, J.Y., Haghanah, V., Kogson-Hurtado, L.M., McAndrew, B.J., Penman, D.J., 2000. Masculinization of genetic female Nile tilapia (*Oreochromis niloticus*) by dietary administration of an aromatase inhibitor during sexual differentiation. *J. Exp. Zool.* 287, 46–53.
- Kwon, J.Y., Mc Andrew, B.J., Penman, D.J., 2001. Cloning of brain aromatase gene and expression of brain and ovarian aromatase genes during sexual differentiation in genetic male and female Nile tilapia *Oreochromis niloticus*. *Mol. Reprod. Dev.* 59, 359–370.
- Lee, B.Y., Kocher, T.D., 2007. Exclusion of Wilms tumor (WT1) and ovarian cytochrome P450 aromatase (CYP19A1) as candidates for sex determination genes in Nile tilapia (*Oreochromis niloticus*). *Anim. Genet.* 38, 85–86.
- Lee, B.Y., Penman, D.J., Kocher, T.D., 2003. Identification of a sex-determining region in Nile tilapia (*Oreochromis niloticus*) using bulked segregant analysis. *Anim. Genet.* 34, 379–383.
- Lee, B.Y., Hulata, G., Kocher, T.D., 2004. Two unlinked loci controlling the sex of blue tilapia (*Oreochromis aureus*). *Heredity* 92, 543–549.
- Löffler, K.A., Zarkower, D., Koopman, P., 2003. Etiology of ovarian failure in blepharophthalmos ptosis epicanthus inversus syndrome: FOXL2 is a conserved, early-acting gene in vertebrate ovarian development. *Endocrinology* 144, 3237–3243.
- Mair, G.C., Scott, A.G., Penman, D., Beardmore, J.A., Skibinski, D.O., 1991a. Sex determination in genus *Oreochromis*: 1—sex reversal, gynogenesis and triploidy in *O. niloticus* (L.). *Theor. Appl. Genet.* 82, 144–152.
- Mair, G.C., Scott, A., Penman, D.J., Skibinski, D.O.F., Beardmore, J.A., 1991b. II. Sex reversal, hybridisation, gynogenesis and triploidy in *O. aureus* Steindachner. *Theor. Appl. Genet.* 82, 153–160.
- Mair, G.C., Abucay, J.S., Skibinski, D.O.F., Abella, T.A., Beardmore, J.A., 1997. Genetic manipulation of sex ratio for the large-scale production of all-male tilapia, *Oreochromis niloticus*. *Can. J. Fish. Aquat. Sci.* 54, 396–404.
- Majumdar, K.C., McAndrew, B.J., 1986. Relative DNA content of somatic nuclei and chromosomal studies in three genera: Tilapia, *Sarotherodon* and *Oreochromis* of the tribe Tilapiini. *Genetica* 68, 165–168.
- Martins, C., Oliveira, C., Wasko, A.P., Wright, J.M., 2004. Physical mapping of the Nile tilapia (*Oreochromis niloticus*) genome by fluorescent in situ hybridization of repetitive DNAs to metaphase chromosomes—a review. *Aquaculture* 231, 37–49.
- Matsuda, M., Nagahama, Y., Shinomiya, A., Sato, T., Matsuda, C., Kobayashi, T., Morrey, C.E., Shibata, N., Asakawa, S., Shimizu, N., Horik, H., Hamaguchi, S., Sakaizumi, M., 2002. DMY is a Y-specific DM-domain gene required for male development in the medaka fish. *Nature* 417, 559–563.
- Miyamoto, Y., Taniguchi, H., Hamel, F., Silversides, D.W., Viger, R.S., 2008. A GATA4/WT1 cooperation regulates transcription of genes required for mammalian sex determination and differentiation. *BMC Mol. Biol.* 9, 44.
- Morrison, C.M., Miyake, T., Wright Jr, J.R., 2001. Histological study of the development of the embryo and early larva of *Oreochromis niloticus* (Pisces: Cichlidae). *J. Morphol.* 247, 172–195.
- Müller-Belecke, A., Hörstgen-Schwark, G., 1995. Sex determination in tilapia (*Oreochromis niloticus*): sex ratios in homozygous gynogenetic progeny and their offspring. *Aquaculture* 137, 57–65.
- Nakamura, M., 1975. Dosage-dependent changes in the effect of oral administration on gonadal sex differentiation in *Tilapia mossambica*. *Bull. Fac. Fish Hokkaido Univ.* 26, 99–108.
- Nakamura, M., Kobayashi, T., Chang, X.T., Nagahama, Y., 1998. Gonadal sex differentiation in teleost fish. *J. Exp. Zool.* 281, 362–372.
- Nanda, I., Kondo, M., Hornung, U., Asakawa, S., Winkler, C., Shimizu, A., Shan, Z., Haaf, T., Shimizu, N., Shima, A., Schmid, M., Scharf, M., 2002. A duplicated copy of DMRT1 in the sex-determining region of the Y chromosome of the medaka, *Oryzias latipes*. *Proc. Natl. Acad. Sci. USA* 99, 11778–11783.
- Ohno, S., 1967. *Sex Chromosomes and Sex-Linked Genes*. Springer-Verlag, Berlin.
- Ospina-Alvarez, N., Piferrer, F., 2008. Temperature-dependent sex determination in fish revisited: prevalence, a single sex ratio response pattern, and possible effects of climate change. *PLoS ONE* 3 (7), e2837.
- Philippart, J.C., Ruwet, J.C., 1982. Ecology and distribution of tilapias. In: Pullin, R.S.V., Lowe-McConnell, R.H. (Eds.), *The Biology and Culture of Tilapias*, vol. 7. ICLARM, Manila, Philippines, pp. 15–60.
- Pieau, C., 1971. Sur la proportion sexuelle chez les embryons de deux Chéloniens (*Testudo graeca* L. et *Emys orbicularis* L.) issus d'oeufs incubés artificiellement. *C. R. Acad. Sci. Paris* 272, 3071–3074.
- Pieau, C., 1972. Effets de la température sur le développement des glandes génitales chez les embryons de deux Chéloniens, *Emys orbicularis* L. et *Testudo graeca* L. *C. R. Acad. Sci. Paris* 274, 719–722.
- Quinn, A.E., Georges, A., Sarre, S., Guarino, F., Ezaz, T., Marshall Graves, J.A., 2007. Temperature sex reversal implies sex gene dosage in a reptile. *Science* 316, 411.
- Radder, R.S., Quinn, A.E., Georges, A., Sarre, S.D., Shine, R., 2008. Genetic evidence for co-occurrence of chromosomal and thermal sex-determining systems in a lizard. *Biol. Letts.* 4, 176–178.
- Rougeot, C., Kanfitine, S.Y., Prignon, C., Gennotte, V., Mèlard, C., 2007. Early sex reversal during the embryonic development in the Nile tilapia. In: Roudaut, G., Labbé, C., Bobe, J. (Eds.), *Proceedings of the 8th International Symposium on Reproductive Physiology of Fish*, June 3–8, Saint-Malo, France, p. 161.
- Rougeot, C., Prignon, C., Ngouana Kenge, C.V., Mèlard, C., 2008. Effect of high temperature during embryogenesis on the sex differentiation process in the Nile tilapia, *Oreochromis niloticus*. *Aquaculture* 276, 205–208.

- Sarre, S.D., Georges, A., Quinn, A., 2004. The ends of a continuum: genetic and temperature-dependent sex determination in reptiles. *Bioessays* 26, 639–645.
- Scott, A.G., Penman, D.J., Beardmore, J.A., Skibinski, D.O.F., 1989. The 'YY' supermale in *Oreochromis niloticus* (L.) and its potential in aquaculture. *Aquaculture* 78, 237–251.
- Shine, R., Elphick, M.J., Donnellan, S., 2002. Co-occurrence of multiple, supposedly incompatible modes of sex determination in a lizard population. *Ecol. Lett.* 5, 486–489.
- Shirak, A., Seroussi, E., Cnaani, A., Howe, A.E., Domokhovskiy, R., Zilberman, N., Kocher, T.D., Hulata, G., Ron, M., 2006. *Amh* and *dmrt2* genes map to tilapia (*Oreochromis* spp.) linkage group 23 within quantitative trait locus regions for sex determination. *Genetics* 174, 1573–1581.
- Shoemaker, C., Ramsey, M., Queen, J., Crews, D., 2007. Expression of Sox9, Mis, and Dmrt1 in the gonad of a species with temperature-dependent sex determination. *Dev. Dyn.* 236, 1055–1063.
- Smith, C.A., Sinclair, A.H., 2004. Sex determination: insights from the chicken. *Bioessays* 26, 120–132.
- Strüssmann, C.A., Patiño, R., 1999. Sex determination, environmental. In: Knobil, E., Neill, J.D. (Eds.), *Encyclopedia of Reproduction*. Academic Press, New York, pp. 402–409.
- Strüssmann, C.A., Moriyama, S., Hanke, E.F., Calsina Cota, J.C., Takashima, F., 1996. Evidence of thermolabile sex determination in pejerrey. *J. Fish Biol.* 48, 643–651.
- Tessema, M., Müller-Belecke, A., Hörstgen-Schwark, G., 2006. Effect of rearing temperatures on the sex ratios of *Oreochromis niloticus* populations. *Aquaculture* 258, 270–277.
- Toguyeni, A., Fauconneau, B., Fostier, A., Abucay, J., Mair, G.C., Baroiller, J.F., 2002. Influence of sexual phenotype and genotype, and sex-ratio on growth performances in tilapia, *Oreochromis niloticus*. *Aquaculture* 207, 249–261.
- Trewavas, E., 1983. Tilapiine Fishes of the Genera *Sarotherodon*, *Oreochromis* and *Danakilia*. British Museum Natural History, London, UK.
- Valenzuela, N., 2004. Evolution and maintenance of temperature-dependent sex determination. In: Valenzuela, N., Lance, V. (Eds.), *Temperature-Dependent Sex Determination in Vertebrates*. Smithsonian Books, Washington, DC, pp. 131–147.
- Valenzuela, N., 2008. Sexual development and the evolution of sex determination. *Sex. Dev.* 2, 64–72.
- Wang, L.H., Tsai, C.L., 2000. Effects of temperature on the deformity and sex differentiation of tilapia, *Oreochromis mossambicus*. *J. Exp. Zool.* 286, 534–537.
- Wang, D.S., Kobayashi, T., Zhou, L.Y., Paul-Prasanth, B., Ijiri, S., Sakai, F., Okubo, K., Morohashi, K., Nagahama, Y., 2007. Foxl2 up-regulates aromatase gene transcription in a female-specific manner by binding to the promoter as well as interacting with ad4 binding protein/steroidogenic factor 1. *Mol. Endocrinol.* 21, 712–725.
- Warner, D.A., Shine, R., 2008. The adaptive significance of temperature-dependent sex determination in a reptile. *Nature* 451, 566–568.
- Wessels, S., Hörstgen-Schwark, G., 2007. Selection experiments to increase the proportion of males in Nile tilapia (*Oreochromis niloticus*) by means of temperature treatment. *Aquaculture* 272, S80–S87.
- Wessels, S., Hörstgen-Schwark, G., 2008. The influence of rearing temperature and spawner's temperature responsiveness on progeny sex ratios in matings with Nile tilapia YY males. *Proceedings of the 32nd Annual Larval Fish Conference*, August 4–7, Kiel, Germany.
- Western, P.S., Harry, J.L., Graves, J.A., Sinclair, A.H., 1999. Temperature-dependent sex determination in the American alligator: AMH precedes SOX9 expression. *Dev. Dyn.* 216, 411–419.
- Yamamoto, T., 1969. Sex differentiation. *Fish Physiol.* 3, 117–175.
- Yao, H.H., Capel, B., 2005. Temperature, genes, and sex: a comparative view of sex determination in *Trachemys scripta* and *Mus musculus*. *J. Biochem.* 138, 5–12.